

¹²⁵I episcleral plaque brachytherapy in the treatment of choroidal melanoma: A single-institution experience in Spain

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ABSTRACT

PURPOSE: To analyze the results of episcleral plaque brachytherapy from the Catalan Institute of Oncology in Spain.

MATERIALS AND METHODS: From September 1996 through December 2004, 120 patients with choroidal melanoma (median age, 59 years) were treated with iodine-125 seeds at our institution. Patients were classified according to the criteria developed by the Collaborative Ocular Melanoma Study (COMS) group, as follows—COMS-I: 3 patients; COMS-II: 87 patients; COMS-III: 24 patients; and indeterminate COMS: 9 patients. Followup ranged from 1 year to 8.4 years.

RESULTS: Overall survival at 5 and 8 years was 83.9% and 73.3%, respectively. The 5- and 8-year specific survival rate was 85.7%. Local control was 88.2% and 72.7% at 5 and 8 years, respectively. The most common treatment-related toxicity was cataract formation (31.6% of cases), followed by radiation retinopathy (7.5%) and retinal detachment (4.1%).

CONCLUSION: The results of this institutional retrospective study confirm that the use of iodine-125 episcleral plaques to treat choroidal melanoma offers the potential for conserving a functioning eyeball. The toxicity profile is favorable and disease control is similar to other techniques. © 2009 American Brachytherapy Society. Published by Elsevier Inc. All rights reserved.

Keywords:

Choroidal melanoma; Brachytherapy; Episcleral plaque; COMS; ¹²⁵I

Introduction

Enucleation has been the standard treatment for choroidal melanoma since the end of the 19th century; however, its effect on survival has not been well documented (1, 2). The desire to improve survival while preserving visual function and cosmesis in choroidal melanoma patients has led to the rise of alternative treatments, most notably brachytherapy. However, numerous conflicting reports on the value of brachytherapy were published in the 1980s, with some studies

finding brachytherapy to be worse than enucleation (3), others reporting better results, and yet others finding brachytherapy to be equally effective in terms of survival (4). In the United States, this controversy led to the creation of the Collaborative Ocular Melanoma Study (COMS) group, which conducted two multicenter randomized clinical trials for large and medium choroidal melanomas, respectively, and an observational study to evaluate small choroidal melanomas.

The COMS study of patients with large choroidal melanomas found no significant difference in risk of death between patients treated with pre-enucleation radiation compared with those who underwent enucleation. Moreover, estimated 5-year survival rates for patients with histologically confirmed metastatic melanoma were virtually identical (72% for enucleation alone vs. 74% for pre-enucleation radiation) (5). In the second study, 1317 patients with medium choroidal melanomas were randomized to enucleation or

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brachytherapy with iodine-125 (^{125}I). No significant difference in overall survival (OS) rate ($p = 0.48$) was observed. The adjusted estimated risk ratio for brachytherapy vs. enucleation was 0.9 (95% confidence interval [95% CI]: 0.8–1.2). The 5-year death rate for histologically confirmed metastasis was 11% after enucleation and 9% after brachytherapy. The estimated risk ratio was 0.91 (95% CI: 0.6–1.2) (6). In the observational study, 204 patients with small choroidal melanoma were followed clinically. Of these, 75 underwent treatment during the enrollment or followup periods. The 5- and 8-year tumor-specific mortality rates were 1% (95% CI: 0–2.5%) and 3.7% (95% CI: 0.7–6.6%), respectively. The study concluded that healthy 60-year-old patients with small choroidal melanoma had a low risk of dying within 5 years (7).

The COMS trials performed to date have shown that preoperative radiotherapy for large melanomas does not improve survival compared with enucleation alone. Furthermore, survival rates in medium-sized melanomas are similar whether patients are treated by brachytherapy or enucleation. The results of these studies show that brachytherapy is a good alternative to enucleation.

Given the findings of the COMS trials, we decided to perform a retrospective review at our institution to evaluate our results with this procedure.

Methods and materials

Our group retrospectively reviewed records from 120 patients with choroidal melanoma diagnosed at the Bellvitge Hospital in Spain and treated at the Catalan Institute of Oncology between September 1996 and December 2004. All patients signed an informed consent form after being duly informed about possible side effects. Local control (LC), OS, metastasis-free survival (MFS), treatment-related toxicity, and need for enucleation were assessed.

All patients underwent a complete ophthalmoscopic study to assess the clinical characteristics of the tumor. A and B-mode ultrasound with the Ophthascan S echograph (Alcon Laboratories, Fort Worth, TX) was used to confirm the clinical diagnosis and to measure the tumor dimensions (height, horizontal and vertical diameter). All patients underwent retinography during the initial visit and at all subsequent consultations to monitor the local effect of treatment on tumor progression. A radiation oncologist performed the physical examination. Systemic tests (general laboratory tests, complete blood count, liver markers, chest X-ray, and liver ultrasound) were carried out to rule out metastatic involvement.

Based on the COMS classification system, brachytherapy was indicated in the following cases—small melanomas with a documented tendency to grow or that presented clear signs of activity at the initial visit; all medium-sized melanomas; some large melanomas (in cases with a reasonable potential for visual conservation,

assuming the availability of adequate plaques to provide good safety margins and patient agreement) (Table 1).

Local failure was established by clinical determination of tumor expansion or extrascleral extension greater than 2 mm after brachytherapy. Although a decrease in tumor height was observed even up to 6 months after brachytherapy, we followed the criteria established by COMS for local failure—initial increase of at least 15% in the height as measured by ultrasound, or an expansion of 0.25 mm of any tumor dimension as measured by photography or ophthalmoscopic examination.

To perform the three-dimensional reconstruction of the eyeball and the tumor (tumor height, width, and length), as well as the topographic relationship with organs at risk, the clinical ophthalmologic examination was transferred to a predefined graphical scheme and then digitized into a computer system developed by M. Astraham at the University of California (BEBIG Plaque Simulator, version 2.16). Before treatment, we determined the following information—(1) plaque size, number, activity, and distribution of ^{125}I seeds required to provide the prescribed dose to the target volume, (2) doses to sclera, lens, optic nerve, and macula, and (3) treatment length needed to reach the prescribed dose.

Implant sequence

Patient preparation

Insertion of the radioactive plaque was carried out with both local and general anesthetic (retrobulbar anesthesia with akinesis in most cases).

Surgical maneuvers

A limbic peritomy was performed, dissecting Tenon's capsule to expose the scleral surface with the tumor base. If the lesion was near the muscle insertion area, the muscle was temporarily removed.

Clinical target volume determination

Pupillary transillumination was used for localization and marking of the base of the melanoma. Various nonabsorbable sutures were placed at the scleral level, at the positions marked by a dummy plaque, to assure correct plaque size; the definitive radioactive plaque was then positioned and the sutures tied. The diameter of the radioactive applicator

Table 1
COMS classification

COMS	Height	Basal diameter
Small (mm)	≤ 2.5	< 10
Medium (mm)	2.5–10	< 16
Large (mm)	> 10	> 16
Metastatic	N1 or M1	

COMS = Collaborative Ocular Melanoma Study.

was 2 mm larger than the tumor base to treat presumed microscopic disease extension.

Postoperative treatment

Anti-inflammatory drugs and antibiotics were administered subconjunctivally, topically, and systemically in the operating room. The patient was subsequently transferred to a shielded room.

Treatment course

Treatment was given as scheduled and all patients remained in the Brachytherapy Unit. During admission, analgesic and prophylactic antibiotic treatments were prescribed.

Implant removal

After administration of the prescribed treatment, the patient was again transferred to the implant room for implant removal under anesthesia. Complete removal of the radioactive load was verified. Once the plaque had been extracted, the patient was returned to his or her room until recovery and discharge.

Followup

Patients were monitored by the Radiotherapy and Ophthalmology Departments every 3 months for the first year and every 6 months for the following 5 years, and thereafter annually if no evidence of relapse was found. Chest X-ray, complete blood laboratory tests with liver function tests, and abdominal ultrasound were requested as complementary studies, in addition to a physical examination by the radiation oncologist. An ophthalmologist examined the eye to monitor tumor progression, detect early recurrence, and evaluate acute and chronic toxicity.

Statistical method

The “R” statistical software package (the R Foundation for Statistical Computing, Vienna, Austria) was used for all calculations and a descriptive analysis of quantitative and qualitative variables was performed. Mean, median, 95% CI, and range (maximum and minimum value) were calculated for quantitative variables. For qualitative variables, frequency of occurrence for the various categories was measured. The Kaplan–Meier method was used to determine OS, disease-specific survival (DSS), recurrence-free survival, LC, disease-free survival (DFS), MFS, and enucleation-free survival. Log-rank tests were performed to evaluate differences in the survival curves as a function of COMS stage. Cox regression analysis was used to evaluate the association of the prognostic factors and the estimation of relative risks in OS and in disease control, verifying the proportional hazards hypothesis with a test in each case. To assess toxicity, a descriptive analysis was

performed and logistic regression analysis was performed to check for association between factors.

Results

General description

From September 1996 through December 2004, 120 cases of choroidal melanoma were treated with ophthalmic brachytherapy with ^{125}I seeds. There were no cases of amelanocytic melanoma. The median age was 59 years (25–83 years). Men accounted for 41.6% of the series and women 58.3%. Sixty-one (50.8%) of the tumors were located in the right eye and 59 (49.1%) in the left. Table 2 shows tumor characteristics and distribution of patients by stage. Dosimetric data are provided in Tables 3 and 4. The mean dose rate was 0.76 Gy/h (0.38–1.33 Gy/h) and the median dose to the apex was 83.06 Gy (63.93–106.60 Gy). COMS plaque sizes 10–22 were used, with the most common being COMS 18 and 20.

Survival study

Median followup was 50.4 months (3.2–100.7), with 104 patients alive at the time of the present review—of these survivors, 89 were disease-free, 7 had local recurrence only, 7 presented systemic disease, and 1 had both systemic and local relapse. Twelve deaths were attributable to cancer, and 4 were due to unrelated causes. Of the 16 patients who died, 7 developed a second cancer (none of which was located in the treatment area).

Survival rates at 2 and 5 years, respectively, were as follows—OS, 94.4% (95% CI: 90.1–98.8%) and 84.1% (95% CI: 76.5–92.5%); DSS, 95% (95% CI: 91.1–99.4%) and 85.9% (95% CI: 78.5–94.0%); MFS, 90.5% (95% CI: 85.1–95.3%) and 79.7% (95% CI: 71.5–88.9%); and DFS, 87.8% (95% CI: 81.9–94.3%) and 71.2% (95% CI: 61.8–82.1%). LC was 95.3% (95% CI: 91.5–99.4%) at 2 years and 88.4% (95% CI: 81.0–95.5%) at 5 years.

By stage, overall 2- and 5-year survival rates were as follows—COMS-I: 100% and 100%; COMS-II: 94.9% and 83.6%; and COMS-III: 89.9% and 78.3%. These

Table 2
Tumors and patients characteristics

	N	Min	Median	Max	SD	%
Distance to the optic disc (mm)	120	0	5.5	22	25.9	
Distance to the fovea (mm)	120	1.5	6	42	44.5	
Tumor thickness (mm)	120	0.5	5.9	11	5.7	
Horizontal diameter (mm)	120	1	12.2	20	11.4	
Vertical diameter (mm)	120	1	12	22	11.8	
Tumoral volume (cc)	120	0	407.9	1640	1267	
COMS-I	5					4
COMS-II	87					73
COMS-III	24					20
COMS IND	4					3

N = population; Min = minimum; Max = maximum; SD = standard deviation; COMS = Collaborative Ocular Melanoma Study; IND = indeterminate.

Table 3
Dosimetric characteristics 1

	N	Min	Median	Mean	Max	SD
Dose to apex	120	63.9	83	82.9	106.6	7.21
Dose to macula (Gy)	120	2.6	41.2	56.4	227.8	42.1
Dose to optic nerve (Gy)	120	2.7	36.4	44.7	134.1	28.8
Dose to opposite retina (Gy)	120	2.4	8.9	9.4	19.8	4
Dose to lens (Gy)	120	4.4	23.9	30.2	104.1	20.8
Dose to sclera (Gy)	120	155.7	297	321.5	692.8	116

N = population; Min = minimum; Max = maximum; SD = standard deviation.

differences were not significant ($p = 0.716$). The respective 2- and 5-year LC rates were as follows—for COMS-I: 100% and 100%; COMS-II: 100% and 96.6%; and COMS-III: 81.0% and 73.6%. These differences were statistically significant ($p = 0.000228$) (Fig. 1). The 2 and 5-year DFS rates were as follows—COMS-I: 100% and 100%; COMS-II: 93.4% and 75.6%; and COMS-III: 66.6% and 60.5% (Fig. 2). Finally, the respective 2- and 5-year MFS was as follows—COMS-I, 100% and 100%; COMS-II: 93.4% and 78.3%; and COMS-III: 75.6% and 75.6%. These differences were not significant ($p = 0.326$).

Study of independent factors

We used the log-likelihood ratio test to identify factors that might influence the survival. The following variables were evaluated by sex, age, dose to the tumor apex, dose to the sclera, tumor volume, and COMS stage. None of these variables was found to significantly impact survival, probably because of the small sample size of our series. However, we found a nonsignificant association between higher doses to the apex and improved disease control. Additionally—as expected—a higher COMS stage resulted in worse LC ($p = 0.0331$), with a relative risk of 2.4.

Toxicity study

The frequency and severity of treatment-related toxicity was associated with the dose to the eye structures at risk; no significant toxicity that might have required surgery or hospital admission—such as significant strabismus, keratopathy, endophthalmitis, uveitis, retinal detachment, or optic neuropathy—was found. The most commonly observed toxicities were cataract (31.6%), radiation retinopathy (7.5%), glaucoma (7.5%), and vitreous hemorrhage (3.3%). We found one case (0.8%) of strabismus caused by temporary muscle removal during surgery.

Table 4
Dosimetric characteristics 2

	N	Min	Mean	Max	SD
Activity per seed Ci (Bq)	120	1.4 (5.8×10^{10})	3.5 (12.9×10^{10})	5.5 (20.3×10^{10})	1 (37×10^{10})
Activity of the plaque Ci (Bq)	120	12.6 (46.2×10^{10})	56.8 (210.1×10^{10})	104.7 (387.3×10^{10})	22.5 (83.2×10^{10})
Dose rate (Gy/h)	120	0.3	0.7	13	4.6

N = population; Min = minimum; Max = maximum; SD = standard deviation; Ci = Curies; Bq = Becquerel.

Enucleation

Twelve patients (10%) required enucleation after brachytherapy due to toxicity or disease progression; of these, six operations were due to toxicity (5 cases of glaucoma and 1 retinal detachment) and six due to tumor recurrence. Enucleation-free survival for all causes was 86% at 5 years. Most patients (90%) did not require enucleation for any reason.

Discussion

Survival and tumor control

Our main aim was to assess OS and tumor control to compare our results with those of other similar studies. In terms of OS, our results are in line with those reported by several authors.

Jensen *et al.* recently published data on a series of 156 patients with choroidal melanoma treated with ^{125}I at the Mayo Clinic, reporting 5-year LC rates of 92% and 5- and 10-year survival rates of 91% and 79%, respectively (8). These differences with our study may be explained by difference in the study population. Although the percentage of patients at COMS-II was similar (79% vs. 72.5% in our study), Jensen *et al.* reported more patients at COMS-I (19% vs. 4.1% in our study) and fewer COMS-III patients (2% vs. 20%). Additionally, 3.3% of our patients were cases treated for salvage after resection. Given this context, our poorer results are unsurprising.

Differences with the Jensen study were also seen in terms of LC for COMS-II (91.1% vs. 96.6% in our series) and COMS-III (unreported by Jensen *et al.* vs. 73.6% in our series). Moreover, Jensen *et al.* reported a higher MFS for COMS-II patients (87.2% vs. 74.6%). Jensen also found that survival decreased and the rate of distant metastasis increased as tumor size increased and dose to the apex decreased. In short, because of the difference in study populations, it is difficult to compare our study with Jensen *et al.*, except for COMS-II, with similar results in both series.

De Potter *et al.* retrospectively studied 93 patients (mean followup, 78 months) with juxtapapillary choroidal melanoma treated with brachytherapy and found a local recurrence rate of 15% and a time to local failure of 41 months (9). Similarly, Gunduz *et al.*, in a series of 630 patients, reported a 5-year LC rate of 91% and 5- and 10-year metastasis rates of 12% and 22%, respectively (10). Unlike our study, both of the aforementioned series used an isotope other than ^{125}I .

Table 5
Toxicity

	Retinopathy (%)	Neuropathy (%)	Glaucoma (%)	Vitreous bleed (%)	Cataract (%)	Maculopathy (%)
COMS (15)	N/A	N/A	N/A	N/A	19	N/A
Jensen et al. (8)	7.5	11	N/A	15	19	13
De Potter et al. (9)	87	N/A	N/A	N/A	N/A	N/A
Gunduz et al. (10)	N/A	13	N/A	N/A	35	40
Heindl et al. (12)	13	5	3	N/A	N/A	N/A
Fontanesi et al. (19)	21.5	N/A	N/A	N/A	N/A	N/A
Shields et al. (11)	25	22	21	23	66	24
Packer et al. (20)	23.4	N/A	N/A	N/A	N/A	N/A
Nag et al. (17)	40	N/A	N/A	N/A	N/A	N/A
Moataz et al. (18)	27	8.4	N/A	N/A	31	N/A
ICO	7.5	0.83	7.5	3	31.67	4.7

ICO = Instituto Catalan de Oncologia; COMS = Collaborative Ocular Melanoma Study; N/A = not available.

Shields *et al.* evaluated 354 patients with large-sized melanomas (>8 mm in thickness) treated with ¹²⁵I or ruthenium-106 (¹⁰⁶Ru) and reported local recurrence rates of 9% at 5 years and 13% at 10 years. The authors identified two risk factors for recurrence—the use of Ru plaques and tumors involving the ciliary body. The 5- and 10-year MFS was 60% and 45%, respectively (11).

Heindl *et al.* recently published their experience with 100 patients treated with ¹⁰⁶Ru for choroidal and ciliary body melanoma. The 5-year LC and survival rates were 93% and 91%, respectively. Toxicity rates were as follows—13% retinopathy, 5% optic neuropathy, and 3% glaucoma. Enucleation was necessary in 14% of patients and metastasis occurred in 10% of cases. The authors concluded that ¹⁰⁶Ru provides good results for melanomas smaller than 5 mm, although this isotope is not appropriate for larger melanomas because of its dosimetric characteristics (12).

Damato *et al.* also analyzed a series of patients treated with ¹⁰⁶Ru. A total of 458 patients, with a tumor height ≤5.4 mm and a diameter ≤16 mm, were treated from January 1993 through December 2001, with a median dose of 115 Gy to the tumor and without a safety margin. With a median followup of 3.9 years, the actuarial recurrence at 2, 5, and 7 years was 1%, 2%, and 3%, respectively. The authors found that tumor size was the only prognostic factor for local failure. No survival data were reported (13).

Table 6
Series 1

Series	Isotope	Followup (y)	OS (%)	MFS (%)	DFS (%)	N
Jensen et al. (8)	¹²⁵ I	6.2	80	87.2	91	156
De Potter et al. (9)	¹⁰⁶ Ru	6.5	N/A	N/A	N/A	93
Gunduz et al. (10)	¹⁰⁶ Ru	5.2	N/A	N/A	88	630
Heindl et al. (12)	¹⁰⁶ Ru	N/A	N/A	N/A	91	100
Damato et al. (13)	¹⁰⁶ Ru	3.9	N/A	N/A	N/A	458
Petrovich et al. (14)	¹²⁵ I	37 (mo)	88	89	N/A	85
Nag et al. (17)	¹²⁵ I	4.1	N/A	N/A	86	78
Moataz et al. (18)	¹²⁵ I	5.7	75	N/A	90	119

OS = overall survival; MFS = metastasis-free survival; DFS = disease-free survival; N = population; ¹²⁵I = iodine-125; ¹⁰⁶Ru = ruthenium-106; N/A = not available.

Petrovich *et al.* evaluated a series of 85 patients with COMS-II choroidal melanoma and found an actuarial survival at 5 and 8 years of 88% and 84%, respectively. As in other series, these authors also found that survival and risk of metastatic disease were both related to the tumor size, with survival decreasing and metastasis increasing with larger tumors (14).

Of course, the largest and most important prospective, randomized study was conducted by the COMS group (15). Five years after enrollment, 10.3% of eyes (95% CI: 8.0–13.2%) exhibited treatment failure and the 5-year cumulative risk of enucleation in this cohort was slightly higher—12.5% (95% CI: 10–15.6%). The main reason for enucleation in the first 3 years was local recurrence, although after this period, most enucleations were associated with treatment-related complications, especially pain.

In COMS report number 18 (6), and subsequently in report no. 28 (16), the results of 1317 patients with a 12-year followup were analyzed. COMS Stage II patients were assigned to ¹²⁵I brachytherapy or enucleation. Although the mortality rates for both treatments were similar, the authors emphasized that this result could not be extrapolated to other series given the strict inclusion criteria of this study.

Nag *et al.* also published a series of 78 patients with COMS-II choroidal melanoma treated with ¹²⁵I (median followup, 49 months). Progression-free survival, OS, and DSS at 5 years were 93%, 86%, and 97%, respectively. This study differed from other such studies in the type of plaque used (custom-made, 0.6-mm thick and manufactured in 18k gold) and the dose prescription point—in contrast with the

Table 7
Series 2

Series	Followup (y)	OS (%)	MFS (%)	DFS (%)	LC (%)	N
COMS ¹²⁵ I	1	81	90	82	96.6*	657
ICO ¹²⁵ I	4.2	83.6	74.6	91.63	89.7	120

OS = overall survival; MFS = metastasis-free survival; DFS = disease-free survival; LC = local control; N = population; COMS = Collaborative Ocular Melanoma Study; ¹²⁵I = iodine-125; ICO = Instituto Catalan de Oncologia.

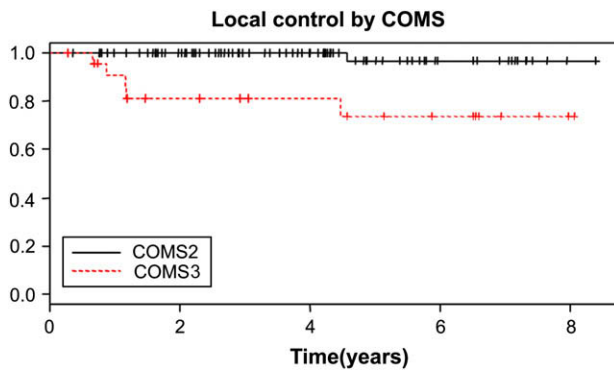


Fig. 1. Local control by Collaborative Ocular Melanoma Study group.

COMS study protocol, in which the prescription point is the tumor apex, Nag *et al.* chose to use the periphery of the tumor at the level of the apex to prevent underdosing at the tumor periphery. Moreover, they also reported that a 1-mm margin (and not 2 mm as recommended by the American Brachytherapy Society) may be sufficient (17).

In October 2006, Moataz *et al.* published a series from Ohio State University that included small- and medium-sized melanomas treated between November 1989 and January 2004. The mean followup was 67 months, with median doses of 85.5 Gy at 5-mm depth, 85 Gy at Nag's prescription point, and 113 Gy at the tumor apex. LC at 5 years was 97.5%. The preservation rate of useful vision was 70%. Progression-free survival was 90% and OS was 75%. Moataz *et al.* found that survival was worse in patients with large tumors (>9 mm) compared with smaller tumors (91% vs. 64%), a finding corroborated in our series. These authors also reported that tumor volume was associated with survival, with significant differences in patients with tumors greater than 200 cc (76% vs. 85%) (18) (Tables 5–7).

Toxicity

Jensen *et al.* found that higher doses to the tumor apex were associated with reduced rates of distant metastases but worse ocular function. In 3% of cases in their series, enucleation was required due to toxicity. Neuropathy,

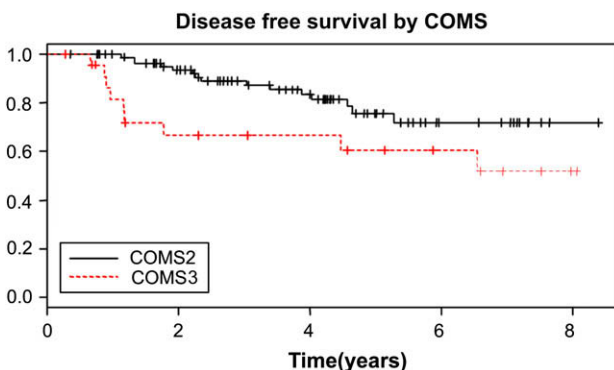


Fig. 2. Disease-free survival by Collaborative Ocular Melanoma Study group.

caused by proximity to the optic nerve, occurred in 11% of patients. Other side effects reported in the Jensen series included cataracts (19% of cases), maculopathy (13%), and vitreous hemorrhage (15%). Rates of retinopathy show wide variation from one study to another. Fontanesi *et al.* (19) reported a 21.5% rate vs. 23.4% described by Packer *et al.* (20) and the remarkably high 87% rate reported by Potter, who also found that diabetes and age were associated with retinopathy. In contrast, the retinopathy rate in our series was only 7.5%. The most commonly occurring toxicity in our case series was cataract, which affected 31.6% of patients. We were unable to find an adequate explanation for these differences.

Gunduz *et al.* reported maculopathy in 40% of cases vs. only 4.1% in our series, and a 13% rate of optical neuropathy vs. only 0.8% in our series. Despite these differences, the percentage of cataracts—at approximately 35%—was very similar to our results, as were rates of enucleation for recurrence or toxicity (11% vs. 10% in our study).

Shields evaluated patients treated with ^{125}I and ^{106}Ru , reporting toxicity as follows—proliferative retinopathy (25% of patients), maculopathy (24%), cataracts (66%), neovascular glaucoma (21%), and vitreous hemorrhage (23%). Importantly, enucleation was necessary in 24% of cases.

Heindl *et al.* also used ^{106}Ru and found a low rate of toxicity with good LC in a selected group of patients whose tumor height did not exceed 5.4 mm. Retinopathy occurred in 13% of patients, optical neuropathy in 5%, and glaucoma in 3%.

The rate of ocular toxicity reported by Damato *et al.* was minimal, perhaps because of the use of transpupillary thermotherapy administered to prevent tumor growth in 41 patients and to treat exudation in 19 patients.

Petrovich *et al.* found higher complication rates for larger tumors, especially those treated before 1988 (67% vs. 35%, $p = 0.010$). The rate of enucleations due to complications was 15%, probably because of the high dose (102.6 Gy to apex tumor). No data on retinopathy, cataracts, or visual acuity was reported in this series.

The COMS-18 report found that the most common complication was loss of visual acuity (visual acuity was better than 20/200 in 37% of irradiated eyes at 5 years posttreatment (21))—we did not evaluate visual acuity—and a 19% rate of cataracts. Enucleations were required in 12% of cases.

Nag *et al.* described a 40% rate of retinopathy, finding that optical neuropathy was more common in those patients with tumors located adjacent to the optical disc. Nag *et al.* also found that tumors next to the optical nerve resulted in worse vision than those at a greater distance (survival with intact vision at 5 years was 50% for tumors located <4 mm from the optic disc compared with 74% for those >4 mm from the optic disc [$p < 0.03$]). Moataz *et al.* found that the most common toxicity profiles were cataract, retinopathy, retinal detachment, maculopathy, and vitreous hemorrhage. He described a 27% rate of retinopathy occurrence, 31% for cataracts—similar to our results—8.4% for papillopathy, and 3% for vitreous hemorrhage.

Our series was partially analyzed by Caminal *et al.* in 2001(22) in a retrospective study that included measurement of visual acuity. After a mean of 2.4 years of follow-up, tumor height and volume decreased by a mean of 53.8% and 75.8%, respectively. As for visual acuity, 32.7% of patients presented a vision of 0.1 or better at the last follow-up consultation, whereas 13.5% presented low vision at diagnosis, a percentage that remained unchanged through the latest followup evaluation.

Conclusion

Ophthalmic brachytherapy with ^{125}I seeds represents an alternative to enucleation and an opportunity to preserve organ and function. Toxicity in our series was moderate, easily manageable, and in most cases did not require admission or invasive treatment. Ninety percent of patients conserved their eyes at 5 years. A higher COMS stage resulted in worse LC. Our results for each tumor stage were similar to those reported by other centers that have used brachytherapy.

Candidates for conservative treatment with brachytherapy are as follows—small-sized choroidal melanoma with documented tendency for growth, all medium-sized melanomas, and some large-sized melanomas (in eyes with certain potential for visual conservation) according to COMS criteria.

The results of our study confirm the findings reported elsewhere. ^{125}I brachytherapy is a good conservative treatment for medium-sized uveal melanoma, and is a reasonable alternative to enucleation, especially considering that brachytherapy makes it possible to preserve eyesight, with good cosmesis.

References

- [1] Zimmerman LE, McLean IW, Foster WD. Does enucleation of the eye containing a malignant melanoma prevent or accelerate the dissemination of tumour cells? *Br J Ophthalmol* 1978;62:420–425.
- [2] Curtis E. The Collaborative Ocular Melanoma Study: An overview. *Cancer Control* 2004;11:304–3091.
- [3] Gass JD. Comparison of prognosis after enucleation vs cobalt 60 irradiation of melanomas. *Arch Ophthalmol* 1985;103:916–923.
- [4] Cruess AF, Augsburger JJ, Shields JA, *et al.* Regression of posterior uveal melanomas following cobalt-60 plaque radiotherapy. *Ophthalmology* 1984;91:1716–1719.
- [5] The Collaborative Ocular Melanoma Study (COMS) randomized trial of pre-enucleation radiation of large choroidal melanoma II: Initial mortality findings. COMS report No. 10. *Am J Ophthalmol* 1998;125:779–796.
- [6] Diener-West M, Earle JD, Fine SL, *et al.* The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma III: Initial mortality findings. COMS report No. 18. *Arch Ophthalmol* 2001;119:969–982.
- [7] The Collaborative Ocular Melanoma Study Group. Factors predictive of growth and treatment of small choroidal melanoma: COMS report No. 5. *Arch Ophthalmol* 1997;115:1537–1544.
- [8] Jensen AW, Petersen IA, Kline RW, *et al.* Radiation complications and tumor control after 125I plaque brachytherapy for ocular melanoma. *Int J Radiat Oncol Biol Phys* 2005;63:101–108.
- [9] De Potter P, Shields CL, Shields JA, *et al.* Plaque radiotherapy for juxtapapillary choroidal melanoma. Visual acuity and survival outcome. *Arch Ophthalmol* 1996;114:1357–1365.
- [10] Gunduz K, Shields CL, Shields JA, *et al.* Radiation complications and tumor control after plaque radiotherapy of choroidal melanoma with macular involvement. *Am J Ophthalmol* 1999;127:579–589.
- [11] Shields CL, Naseripour M, Cater J, *et al.* Plaque radiotherapy for large posterior uveal melanomas (≥ 8 mm thick) in 354 consecutive patients. *Ophthalmology* 2002;109:1838–1849.
- [12] Heindl LM, Lotter M, Strnad V, *et al.* High-dose (106)Ruthenium plaque brachytherapy for posterior uveal melanoma: A clinico-pathologic study. *Ophthalmologie* 2007;104:149–157.
- [13] Damato B, Patel I, Campbell I, *et al.* Local tumor control after ^{106}Ru brachytherapy of choroidal melanoma. *Int J Radiat Oncol Biol Phys* 2005;63:385–391.
- [14] Petrovich Z, Luxton G, Langholz B, *et al.* Episcleral plaque radiotherapy in the treatment of uveal melanomas. *Int J Radiat Oncol Biol Phys* 1992;24:247–251.
- [15] Jampol LM, Moy CS, Murray TG, *et al.* Collaborative Ocular Melanoma Study Group (COMS Group). The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma: IV. Local treatment failure and enucleation in the first 5 years after brachytherapy. COMS report no. 19. *Ophthalmology* 2002;109:2197–2206.
- [16] Collaborative Ocular Melanoma Study Group. The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma: V. Twelve-year mortality rates and prognostic factors: COMS report No. 28. *Arch Ophthalmol* 2006;124:1684–1693.
- [17] Nag S, Wang D, Wu H, *et al.* Custom-made “Nag” eye plaques for 125I brachytherapy. *Int J Radiat Oncol Biol Phys* 2003;56:1373–1380.
- [18] Moataz N, El-Ghamry MD, Subir Nag MD, *et al.* Results of primary treatment of small or medium sized uveal melanoma with custom-made Nag iodine-125 eye plaque: The Ohio State University experience with 119 patients. *Brachytherapy* 2006;5:90.
- [19] Fontanesi J, Meyer D, Xu S, *et al.* Treatment of choroidal melanoma with I-125 plaque. *Int J Radiat Oncol Biol Phys* 1993;26:619–623.
- [20] Packer S, Stoller S, Lesser ML, *et al.* Long-term results of iodine 125 irradiation of uveal melanoma. *Ophthalmology* 1992;99:767–774.
- [21] Collaborative Ocular Melanoma Study Group. The Collaborative Ocular Melanoma Study (COMS) randomized trial of I-125 brachytherapy for medium choroidal melanoma. Visual acuity after 3 years. COMS Report No 16. *Ophthalmology* 2001;108:348–366.
- [22] Caminal JM, Quintana M, Pera J, *et al.* Results of iodine-125 radiotherapy in the treatment of uveal melanoma. *Arch Soc Esp Oftalmol* 2002;77:29–38.